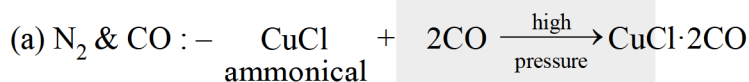


HINTS & SOLUTIONS

Q.1 (B)



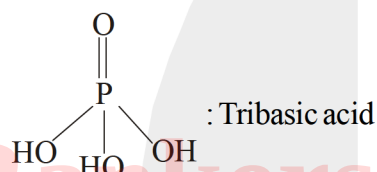
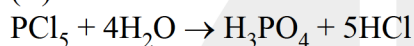
(b) N_2 & O_2 : – O_2 is absorbed in alkaline pyrogallol

(c) N_2 & NH_3 : – $2\text{NH}_3 + \text{H}_2\text{SO}_4 \longrightarrow (\text{NH}_4)_2\text{SO}_4$, while N_2 is neutral gas
base

(d) PH_3 & NH_3 : – NH_3 is highly soluble in water & PH_3 is least soluble in water so can be separated by use of water.



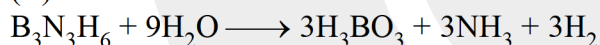
Q.2 (A)



Q.3 (D)

SnCl_4 is stronger Lewis acid than SnCl_2

Q.4 (D)

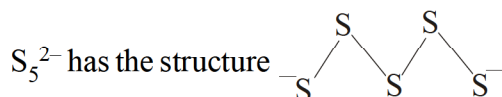
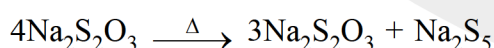
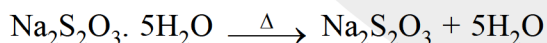


Q.5 (A)

Q.6 (D)

Due to inert pair effect & redox reaction in BiBr_5 due to high polarizing power of Bi^{+5} with Br^-

Q.7 (B)



Q.8 (D)

Due to highest oxidation state.

Q.9 (A)

Q.10 (C)
(C) I does not have its valency 4. It has valency 1, 3, 5 & 7

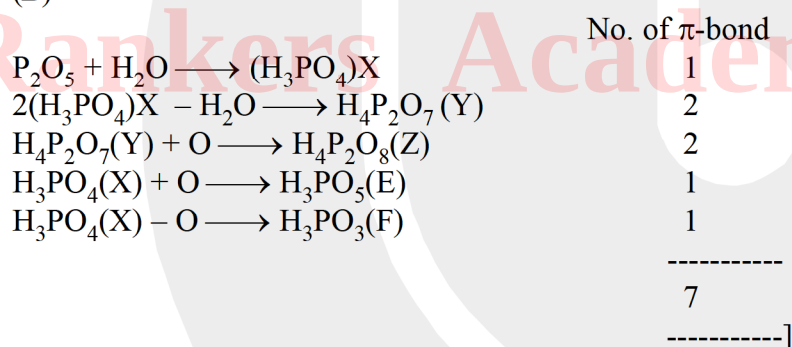
Q.11 (A)
 $\longrightarrow \text{B}_2\text{H}_6 + 2\text{L} \longrightarrow \text{BH}_4^- + [\text{BH}_2\text{L}_2]^+$
 where L may be $\text{NH}_3(\text{CH}_3)\text{NH}_2$ and $(\text{CH}_3)_2\text{NH}$.
 $\longrightarrow \text{B}_2\text{H}_6 + 2\text{L}' \longrightarrow 2(\text{H}_3\text{B} \longleftarrow \text{L}')$
 where L' is ethylenediamine

Q.12 (B)

Q.13 (B)
 Hyponitrous acid exist in dimeric form
 $(\text{HNO}) = \text{H}_2\text{N}_2\text{O}_2$



Q.14 (B)



Q.15 (A)
 $\text{NH}_3 + 3\text{Cl}_2(\text{excess}) \rightarrow \text{NCl}_3 + 3\text{HCl}$
 $\text{NCl}_3 + 3\text{HCl}(\text{aq}) \rightarrow \text{NH}_3(\text{Y}) + 3\text{Cl}_2(\text{Z})$
 $\text{NiCl}_2 + 6\text{NH}_3 \rightarrow [\text{Ni}(\text{NH}_3)_6]\text{Cl}_2$

Q.16 (A)

Q.17 (A)
 $(\text{A}) \rightarrow \text{O} = \text{C} = \text{C} = \text{C} = \text{O}$

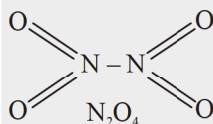
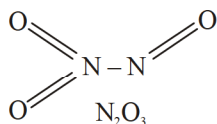
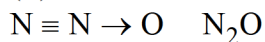
Q.18 (B)
 $\text{Cl}_2\text{O}_6 + 2\text{NaOH} \rightarrow \text{NaClO}_4 + \text{NaClO}_3 + \text{H}_2\text{O}$

Q.19 (D)
 As oxidation state of the central Xe atom increases, its Lewis acid nature increases.

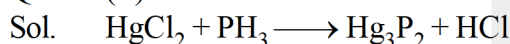
- Q.20 (D)
 (A) $3\text{I}_2 + 6\text{NaOH} \rightarrow 5\text{NaI} + \text{NaIO}_3 + 3\text{H}_2\text{O}$
 (B) $\text{I}_2 + 2\text{Na}_2\text{S}_2\text{O}_3 \rightarrow 2\text{NaI} + \text{Na}_2\text{S}_4\text{O}_6$
 (C) $10\text{HNO}_3 + \text{I}_2 \rightarrow 2\text{HIO}_3 + 10\text{NO}_2 + 4\text{H}_2\text{O}$

Q.21 (C)

Q.22 (B)



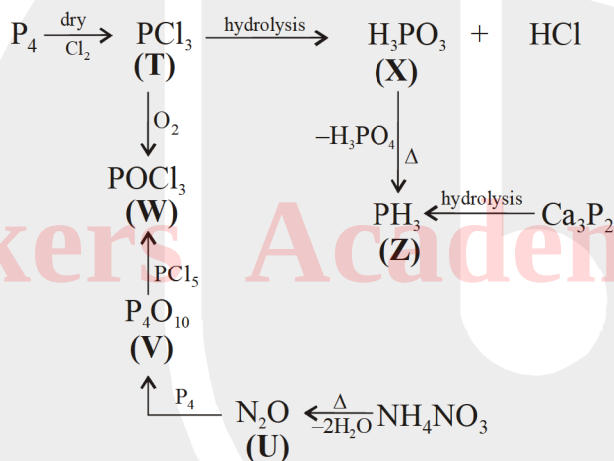
Q.23 (B)



Q.24 (ABC)

Q.25 (ACD)

Q.26 (ABCD)

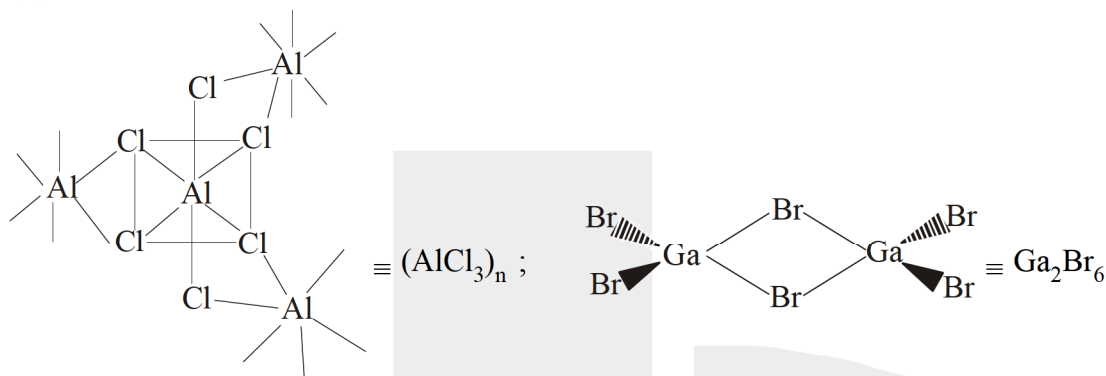


- (i) (A) $2\text{PCl}_5 + \text{H}_2\text{SO}_4 \rightarrow 2\text{POCl}_3 + \text{SO}_2\text{Cl}_2 + 2\text{HCl}$
 (B) $\text{PCl}_5 + \text{SO}_2 \rightarrow \text{POCl}_3 + \text{SOCl}_2$
 (C) $\text{PCl}_5 + \text{H}_2\text{O} \rightarrow \text{POCl}_3 + 2\text{HCl}$
- (ii) (A) $4\text{H}_3\text{PO}_3 \xrightarrow{\Delta} 3\text{H}_3\text{PO}_4 + \text{PH}_3$
 (B) $\text{NH}_4\text{NO}_3 \xrightarrow{\Delta} \text{N}_2\text{O} + 2\text{H}_2\text{O}$
 (C) N_2O supports combustion
 (D) $\text{P}_4 + 3\text{NaOH} + 3\text{H}_2\text{O} \rightarrow 3\text{NaH}_2\text{PO}_2 + \text{PH}_3$
- (iii) $\text{N}_2\text{O} \rightarrow \text{N}_2 + [\text{O}]$

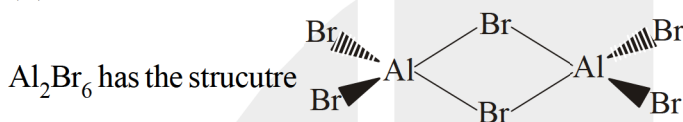
Q.27 (A)

None of the halides of Boron exists as dimer due to ring strain or steric crowding.

Q.28 (D)



Q.29 (B)



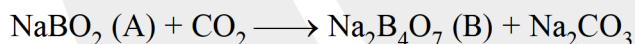
Q.30 (A)

NaBO_3 contains peroxy linkage.

Q.31 (BCD)

Lewis acidic order : BI_3 > BBr_3 > BCl_3 > BF_3
 volatile halide Strong $\text{p}\pi\text{-p}\pi$ back bond
 most thermal stable halide
 $[\text{BI}_4]$ is highly unstable due to steric crowding around boron atom.

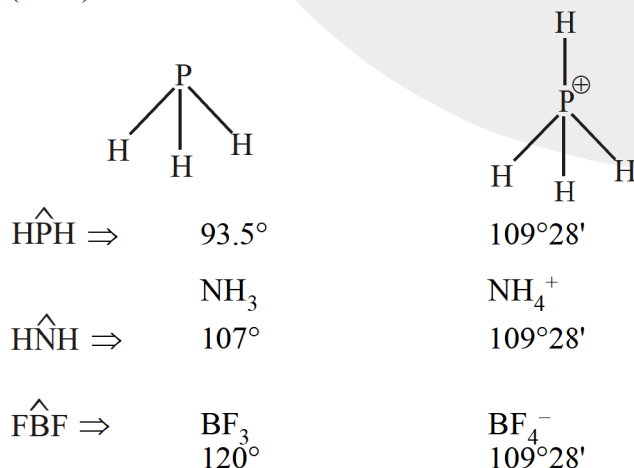
Q.32 (ABD)



Q.33 (AC)

Oxoacids or oxoanions containing intermediate O.S of central atom undergo disproportionation.

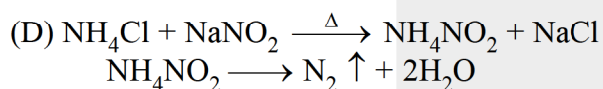
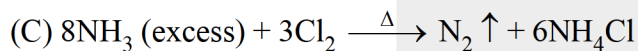
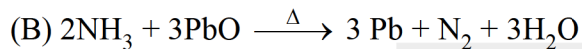
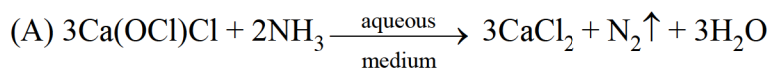
Q.34 (ABC)



Q.35 (ABCD)

Q.36 (BCD)

Q.37 (ABCD)

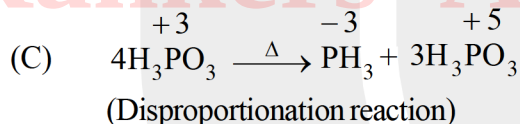
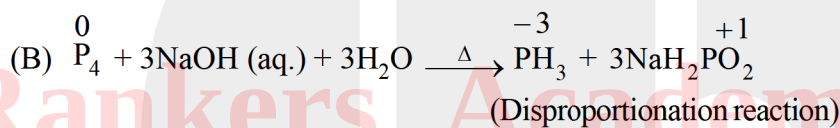
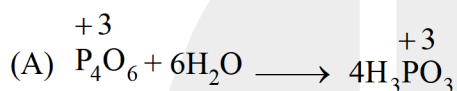


Q.38 (ABD)

NF_3 , CCl_4 and SF_6 compounds are undergo hydrolysis via SN^1 mechanism means in ordinary condition they do not undergo hydrolysis

Q.39 (A) QT (B) PT (C) RT (D) PRS

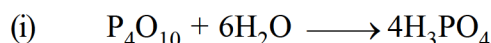
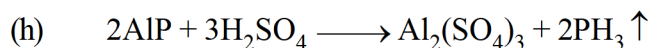
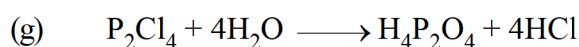
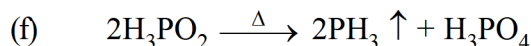
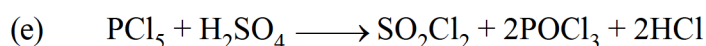
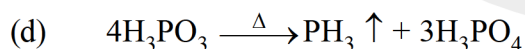
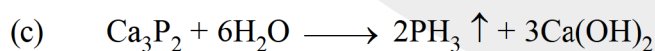
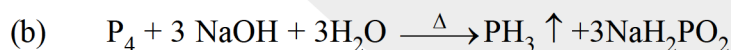
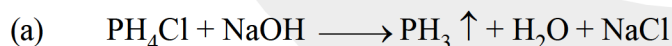
Q.40 (A) P, (B) RS, (C) QRS



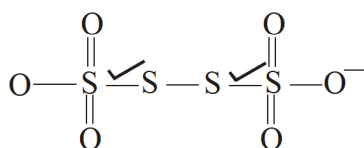
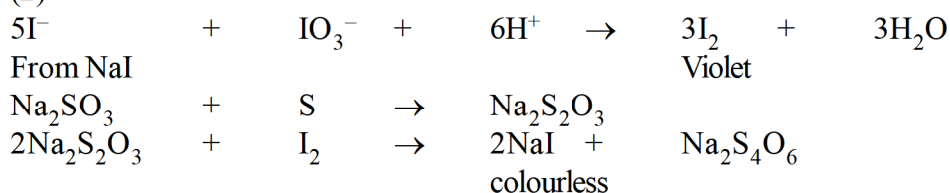
Q.41 (A) s (B) r (C) p (D) q

Q.42 (A) p, s ; (B) q, s (C) r, t ; (D) q, t

Q.43 (0006)

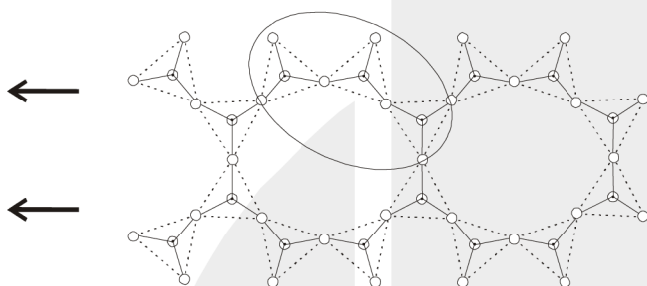


Q.44 (2)



bonds indicated by ✓ are of equal length

Q.45 (7)

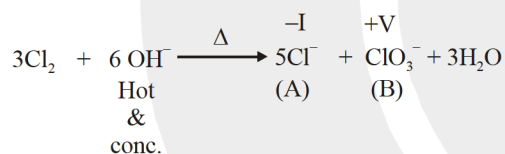


Unit formula = $\text{Si}_3\text{O}_{8.5}^{-5}$ or $\text{Si}_6\text{O}_{17}^{-10}$

$m = 6, n = 17, q = -10$

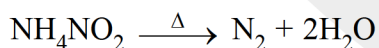
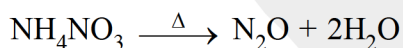
So, $n + q = 17 - 10 = 7$ Ans.

Q.46 (0004)

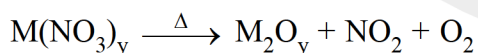


Algebraic sum of oxidation state of Cl in A and B = $-1 + 5 = +4$

Q.47 (0006)



Except $\text{Na}^+, \text{K}^+, \text{Rb}^+, \text{Cs}^+$ all metal nitrate give NO_2 on heating



Q.48 (7)